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It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

21 AI-generated only 60%
Likely AI-generated text from a large-language model.

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Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely AI generated as AI generated and AI paraphrased or likely AI generated and AI paraphrased writing as only AI generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

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How should I interpret Turnitin's AI writing percentage and false positives?
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AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Student Name

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Date of Submission

Discuss the Role of Technologies Like AI, Cloud Computing, and Blockchain in Driving Business Transformation, Using a Real-World Example of a Company That Has Successfully Adapted.

The digital economy has changed the face of business models and made it impossible for businesses to continue as they were, as organizations are now wholly operating, interacting with their customers and managing their resources. Physical products, brock and mortar stores, as well as in person interactions were traditional methods of business. However, with the rise of digital innovation technologies such as Artificial Intelligence (AI), cloud computing, and blockchain, we are in a new era that has revolutionized companies and industries all around the world. This is not the transformation of technology adaptation but a radical reformation of the practices of supplying value, serving customers, and functioning (Akter et al., 2022).

The Digital Economy: A Shift from Physical to Digital

The digital economy uses digital technology to operate its internet-based trading platforms. A digital economy functions differently than standard businesses because it runs in the virtual world through digital services without needing physical infrastructure. Companies now use internet platforms and modern technology to bring together buyers and sellers for easier transaction processes and to offer digital services (Chowdhury 2024).

As an example retail shops and service providers needed to update their way of working when online marketplaces grew popular. Loyalty brands Amazon, Alibaba, and eBay run online platforms to link customers with worldwide product sellers making shopping easier than brick-and-mortar retail stores (G'aybullayev, 2022).

Role of AI in Business Transformation

Among the key forces that are shaping the digital economy, the most powerful of them is Artificial Intelligence (AI). Machine learning, NLP, and computer vision are promising application areas for implementing value-added AI technologies in the public sector, especially in large databases for analysis, decisions, process optimization, and customer satisfaction. AI is incorporated in the society in the most prominently in healthcare, finance, manufacturing, retailing, etc (Oyekunle & Boohene, 2024).

AI is useful to businesses since it takes over operational work that would elsewise have been undertaken by manual means. For instance, many companies have incorporated the use of AI chatbots to deal with customers' questions, reference, and complaints without the need to involve a human operator. Furthermore, artificial intelligence can be employed to estimate the consumers' behavior, propose the related items, and synchronize supply network logistics. Companies such as Amazon have integrated AI in its functioning to deliver suitable products and services to consumers, anticipate the consumers' needs, and manage inventories.

Cloud Computing: A Catalyst for Scalability and Efficiency

Cloud computing is one of another key technology transforming traditional business models. It entities the provision of storage, processing power, database, and many more through the internet with businesses no longer needing to own and maintain physical infrastructure; cloud computing. As a result, this has dramatically reduced the upfront costs of IT infrastructure enabling any business regardless of its size to scale operations and innovate (Kitsios & Kamariotou, 2021).

Organizations are able to have more agility and flexibility in operations with the cloud computing. With no need for large capital investments in hardware, they can expand their infrastructure quickly, both up or down, according to demand. Some of the prominent examples of cloud platforms where business have started adopting a model of scalable and cheaper IT operations are Amazon web services, Microsoft Azure or Google Cloud.

This encourages collaboration, allowing employees and partners to work in unison on these services and tools from any location in the world, boosting productivity and facilitating better decision making. Additionally, cloud computing allows organizations to incorporate sophisticated technologies like AI and machine learning since they no longer need to invest in infrastructure by enabling businesses to access vast computing facilities at their disposal (Arogundade & Palla, 2023).

Blockchain: Revolutionizing Trust and Transparency

One more transformative technology that is disrupting brick and build business models especially in areas that require transparency, security and trust is Blockchain. Blockchain is a distributed ledger technology that works on a decentralized framework that eliminates the requirement of intermediaries to facilitate secure transactions between the parties involved. Each transaction is written on a ‘block,’ and blocks form a ‘chain’ so that the data is immutable and transparent.

Blockchain has one of the primary applications in the digital economy — cryptocurrency. Bitcoin and other cryptocurrencies are decentralized digital currencies which provide an ability to exchange value without depending on the traditional financial institutions. But this isn’t all that blockchain has to offer. The technology can be used to secure supply chains to streamline, and ensure the integrity of digital identities, and improve contract management with smart contracts (Lal et al., 2024).

IBM has successfully enabled the business of its own model by using blockchain technology, particularly in supply chain management. The company enables food industry companies to track the food product journey from farm to table using its IBM Food Trust blockchain platform that increases the transparency and decreases the fraud. However, this blockchain application not only adds a layer of efficiency to the operation but also helps build trust towards consumers who continuously ask for transparency at every step of purchasing products.

Real-World Example: Amazon's Digital Transformation

The successful example of digital transformation comes from Amazon's business performance. Since its start as an online bookstore Amazon has become a worldwide e-commerce leader which applies technologies from AI to blockchain to develop its business. Amazon Web Services transforms IT operations through its affordable cloud solutions that deliver adaptable computing resources to organizations. The AWS business segment delivers strong financial results to Amazon by generating major parts of the total revenue (Lang, 2021). Besides these features Amazon applies AI to improve customer experiences automatically handle fulfillment services and run smoother delivery operations. The company now builds its strategy around Alexa AI and finds new ways to serve smart home customers. Amazon keeps researching blockchain technology while trying to make its supply chain more visible and exploring DeFi alternatives.

Explain How Blockchain, IoT, and Cryptocurrencies Are Influencing IT Management Practices. Provide an Example of a Business That Has Integrated These Technologies to Enhance Efficiency or Innovation.

Blockchain, Internet of Things (IoT), and other Cryptocurrencies have ushered in an era of giving new avenues to the businesses to manage their IT infrastructure and data. Innovations like these are not fads but game changer technologies that are remaking whole industries, lowering operational cost and creating entirely new business models. They make IT management practices rethink data security, connectivity, automation and exchange of values. To conclude this discussion, each of these technologies will be discussed on how they have affected IT management and then give an example of a company that's using these innovations to boost its effectiveness and promote more innovations (Alsharari, 2021).

Blockchain in IT Management

First, Blockchain Technology which is known for its connection with the cryptocurrencies like Bitcoin is a decentralized & distributed ledger that records all transactions through several computers to make it secure and legible. This cuts out the need for intermediaries and allows for direct and safe transactions.

In IT management, the main contribution of blockchain is to the data integrity and security. Traditional databases are also centralized; there is a centralized authority that controls and manages the data. Whereas, blockchain gives a decentralized approach wherein data gets encrypted and is stored across a network of computers (a.k.a nodes). It improves the data security as it becomes nearly impossible for any bad actor to tamper with the data devoid of network detecting (Wang et al., 2022).

One such example of a company that has integrated the blockchain technology into its business model is IBM. IBM's Food Trust blockchain enables businesses in the food industry to trace the journey of products from farm to table with transparency and hardly any food fraud. This makes IT management better by bringing blockchain and IoT device together that allows for monitoring of temperature and humidity etc all through the supply chain in order to create an optimized and transparent system.

Internet of Things (IoT) and IT Management

Internet of Things (IoT) is a network of objects or 'things' which are equipped with sensors, software and technologies to communicate and exchange data over the internet. These are everyday object, such as smart thermostats, wearable health devices, industrial equipment, such as smart meters, and connected vehicles.

This is enabled for example in manufacturing by IoT, in where sensors on machinery detect and measures wear and creates alerts to IT systems until when maintenance can be performed before the failure occurrences. This means machines will operate at its optimum performance, therefore reducing downtime as well as enhance efficiency (Mu & AntwiAfari, 2024).

Automation too is done in businesses with IoT. For example, smart homes use IoT technologies that control lighting, temperature, and security systems, automatically these systems are run based on the user preferences or external factors like weather. In smart factories too, this is seen – machines automatically adjust production schedules, inventory levels and energy usage based on communication between them.

The key example of a company integrating the IoT technology in practice is the General Electric (GE). Predix, GE's platform to connect industrial machines to the internet, and bring real time big data analysis to the manufacturing line, is used to predict and prevent failures to machines. IoT's integration had caused an improvement of operational efficiency, reduction of downtime, and improvement of customer service (Khan et al., 2022).

Cryptocurrencies and IT Management

The digital currencies Bitcoin and Ethereum and Ripple enable users to carry out electronic transactions while performing value transfers within internet-based systems. The blockchain technology enables decentralized cryptocurrency transactions that do not require central banks to validate transactions.

A smart contract is an automated agreement which contains its contractual stipulations coded into programmable computer systems. The contracts work autonomously to perform specified operations when predetermined conditions become satisfied thus making intermediaries unnecessary. IT management experiences automation of process execution which minimizes operational mistakes while speeding transaction speeds at the same time minimizing traditional enforcement expenses (Laurențiu-George, 2022).

The adoption of cryptocurrencies by Tesla stands out as an important case of business implementation. Tesla declared in 2021 that it would allow electric vehicle buyers to use Bitcoin for purchases thus opening new possibilities for Bitcoin utilization by companies in transactions. To accept cryptocurrency payments Tesla needed to build digital payment systems within its information technology structure.

The financial industry faces disruption from DeFi (Decentralized Finance) powered by blockchain technology because this solution enables users to execute borrowing and lending activities and trades without middlemen. The DeFi platforms drive businesses to modify their IT infrastructure because they transform the way financial services operate in the market.

Example of Business Integration: Walmart's Use of Blockchain and IoT

Walmart is a perfect example of a firm that applies both the blockchain and IoT technologies in its operations. Block chain and IoT as a supply chain is used, and works for Walmart in decreasing food borne illness risk.

One of the notable change that walmart has implemented with the help of IBM is the systematic measure on the use of block chains which tracks the food products from farms through supply chain to stores. Temperature and humidity sensors for example help in keeping the correct storage and transport conditions for the food items which is very paramount for their quality and safety.

How Do Traditional IT Disciplines, Such as Business Analysis, IT Governance, and Information Security, Evolve with the Integration of Advanced Technologies Like AI and Quantum Computing? Provide a Brief Case Study to Support Your Explanation.

Business analysis together with IT governance along with information security have experienced profound changes because Artificial Intelligence has merged with quantum computing for business implementations. Technology progress enables the enhancement of traditional IT solutions and generates distinctive hurdles and chance along with necessary demands for both IT experts and organizations. The transformation shapes business approaches to manage IT resources and security while needing them to connect their operational plans with company-wide objectives (How & Cheah, 2024).

Business Analysis in the Age of AI and Quantum Computing

Business analysis is the process through which the business needs are determined and the opportunities are evaluated in relation to IT to come up with solutions that will help improvement of business operations and effectiveness. Originally, business analysts were mainly to write work flow that happens in an organization, identifying the need, wants and gap between the current system and the improved system. Nevertheless, integration of artificial intelligence into the business analysis has affected the wide environs and approach commonly applied.

Using AI in business analysis, aspects relating to predictions concerning trends, customer habits, and potential markets can be made by using data gathered in the past. Machine learning devices can process information in seconds and deliver patterns which otherwise would be rather slow and tedious to accomplish by hand. This helps the analysts to make their recommendations far more effective and efficient through accurate and informed conclusions (How & Cheah, 2023).

And in the future, the analysis of business risks can be widened through using quantum computing, which is the next big thing in analysis for managers of big data outfits. Quantum computers are, in fact, capable of computing at a time and with an efficiency that far outperforms traditional computers. It is possible that business analysts could use this power to find better ways of optimizing supply chain, financing solutions and logistics in what could be much faster methods of doing some of these strategic actions with greater precision. Despite the fact that quantum computing is still in development, the value of such a prospect is immense when it comes to analyzing the data.

Netflix The use of AI in Analysis of a business can be explained through the following example of Netflix. The company recommends to the users interesting content that may include videos, images, and text by analyzing the history and preferences of the user. The use of these models has its basis on artificial intelligence and business anticipations in Netflix hire outlooks to redrew show suggestions, view patterns and potential subscriber addition. Additionally, the incorporation of AI enables netflix to enhance it's offering and customer experience consistent with its status as market leader in the industry.

IT Governance: Ensuring Alignment and Compliance in the Digital Age

IT governance forms a central discipline of a company and it refers to the direction, control and monitoring of organization's IT assets to ensure that these are effectively supporting the business processes of an organization. IT governance in the past was based on the alignment of IT with business processes, managing of risks and adherence to compliance standards.

However, the reincarnation of both IT fields, the AI and the quantum computing, into the business environment has brought additional dimensions to IT governance.

AI and quantum computing both present a new set of capabilities and risks that need to be regulated or governed to make the most of them. For instance, it is sometimes difficult to comprehend how an AI system arrives at a decision, hence why its containers are described as 'black boxes'. This scarcity is detrimental in addressing governance because it becomes the duty of businesses to make sure that AI systems are efficient but also legally responsible and sustainably moral (Ingale, 2024).

IBM's Approach to IT Governance with AI IBM is one of those companies that adopted AI and applied it to the issue of IT governance. Their AI platform; IBM Watson has continued to offer solution for AI systems that enable business organisations to implement the systems with compliance to the set regulations. Simpson describes Watson's governance system that is comprised of rules that define how businesses explain their AI models to the public and having measures that address bias and fairness issues. IBM's case showcases how it is essential for IT governance to adapt for the incorporation of the new technologies for AI while keeping to ethical practices and or adherence to the laws of the land.

Information Security: Protecting Data in the Age of AI and Quantum Computing

Traditionally information security concentrated on protecting data from unauthorized access as well as assuring confidentiality, integrity and availability of information. Nowadays, AI and quantum computing is becoming such a factor, that no matter if we accept them or not, they have already impacted how organizations are looking at security.

AI has raised the ability to identify and respond in real time to security threats. With an intelligence of AI, these security systems analyze network traffic, user behavior, and system logs and pattern match to see if it's a security breach or unusual activity occurring. Then, these systems can automatically do something, like blocking access to the compromised account or alerting the security team that there may be a breach. Today, modern cybersecurity facilities rely heavily on AI powered Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) that can react to the changing threats proactively (Alyami et al., 2022).

In terms of providing a glimpse into the future of information security, Google's quantum computing research is a case study. In 2019, Google announced that it first achieved quantum supremacy where it used a quantum computer to solve a difficult problem that would be millions of years to solve in a classical computer. "The business community has got to start preparing for what quantum computing means to encryption," this discovery reinforces. With the continued development of quantum computers as a reality, Google has been hard at work developing quantum safe cryptography to keep information safe.

How Do Agile Development Practices and Innovation-Driven Initiatives Like Hackathons Support Digital Transformation in Organizations? Discuss the Role of Cloud and Fog Computing in Facilitating This Change.

Modern methodologies, including agile development as well as innovation activities like hackathons, are essential in the process of transformation of organisations. Every business is now shifting to become a digital business to meet the ever-evolving market needs, customers, and technologies. Thus, it becomes quite challenging for business to implement new changes and innovations in to traditional Software Development Lifecycle models. With the help of such principles and methods like agility and related events such as hackathons, companies can be more flexible, cyclical, and collaborative. In addition, the cloud and fog computing have very crucial roles to play in this context that includes the ability to provide the hardware and the framework that would enable the ability to perform efficient and effective fast working, collaboration and most importantly the elastic nature needed to enable various digital change in the organisation.

Agile Development Practices and Digital Transformation

Software development through Agile methods uses a framework that places value on flexibilty together with teamwork and fast delivery of programming updates. Agile methods differ from Waterfall development since they split projects into separate small segments known as sprints instead of following a strict linear program. Sprints create opportunities to enhance performance through continuous development along with quick capability adjustments regarding changing needs. The practice of Agile development supports team-based progress through step-by-step development alongside continuous stakeholder input that enables swift delivery of operating software (Block¸ 2023).

Agile frameworks Scrum along with Kanban serve businesses with established techniques for workflow oversight and task priority organization and product delivery through repeated cycles. The key elements of these frameworks emphasize visibility together with feedback processes that support development of user-focused and goal-orientated software and digital solutions.

Following Agile principles an e-commerce business transforms its operations by adding new digital features to the website and app through small frequent deliveries that make products responsive to customer needs and feedback. Organizations gain market superiority in digital economies through Agile development which enables teams to generate accurate solutions at accelerated speeds.

Hackathons: Driving Innovation and Collaboration

The modern-day society has adopted the culture of using hackathons to create positive changes and foster innovation in organizations. A hackathon can be a rather brief process that will entail the understanding of developers, designers and other persons of the particular problem or the development of new ideas and solutions with the maximum level of cooperation. Hackathons are ideas to breed out of the box, overriding the set traditions and to foster cross-functional teamwork which are relevant with the digital transformation imperative (Kamariotou & Kitsios, 2022).

Techno-creatives are particularly renewed by hackathons since it enhances a spirit of togetherness and essence of learning that goes hand in hand with any digital transformation agenda. In invinting employees of different branch (For instance; marketing or information technology or operation), Hackathon encourage cross polining thus coming up with broad solutions to particular organizational concern.

Another example of hackathons for implementing digital solutions is provided by Facebook. Employees periodically are given freedom to engage in hackathons within the company to come up with any idea that might be of worth in the market. These initiated have led to the creation of products like the Facebook Messenger and Timeline services which are core to the Facebook platform's usage.

The Role of Cloud and Fog Computing in Facilitating Digital Transformation

The success of Agile development practices and hackathons relies heavily on the technological infrastructure that supports **collaboration**, **scalability**, and **efficiency**. This is where **cloud computing** and **fog computing** come into play, offering the necessary resources to facilitate agile workflows and innovation-driven initiatives.

Agile teams can leverage cloud platforms such as **Amazon Web Services (AWS)**, **Microsoft Azure**, or **Google Cloud** to quickly spin up virtual environments for development, testing, and deployment. These cloud platforms provide a range of tools, including development frameworks, databases, machine learning models, and AI services, which can be easily integrated into Agile workflows (Kuchuk & Malokhvii, 2024). The cloud also supports the **DevOps** approach, where development and operations teams collaborate to automate the software delivery pipeline and ensure continuous integration and continuous deployment (CI/CD).

For instance, in a manufacturing environment, **IoT sensors** installed on production equipment generate data that needs to be processed quickly to prevent equipment failure or optimize operations. By processing data at the edge through fog computing, organizations can respond to these events in real-time without the delay of sending data to a centralized cloud server.

A key example of fog computing in action is **Cisco**, which integrates fog computing into its **IoT solutions** to improve industrial automation, smart cities, and healthcare systems. Cisco's fog computing infrastructure allows businesses to collect, analyze, and act on data in real-time, enabling faster decision-making and more efficient operations.

References

Akter, S., Michael, K., Uddin, M. R., McCarthy, G., & Rahman, M. (2022). Transforming business using digital innovations: The application of AI, blockchain, cloud and data analytics. *Annals of Operations Research*, 1-33. https://doi.org/10.1007/s10479-020-03620-w

Chowdhury, R. H. (2024). The evolution of business operations: unleashing the potential of Artificial Intelligence, Machine Learning, and Blockchain. *World Journal of Advanced Research and Reviews*, *22*(3), 2135-2147.https://doi.org/10.30574/wjarr.2024.22.3.1992

G'aybullayev, S. (2022). The place of the digital economy today. *Journal of Academic Research and Trends in Educational Sciences*, *1*(11), 116-126. http://ijournal.uz/index.php/jartes

Oyekunle, D., & Boohene, D. (2024). Digital transformation potential: The role of artificial intelligence in business. *International Journal of Professional Business Review: Int. J. Prof. Bus. Rev.*, *9*(3), 1.

Kitsios, F., & Kamariotou, M. (2021). Artificial intelligence and business strategy towards digital transformation: A research agenda. *Sustainability*, *13*(4), 2025.**https://doi.org/10.3390/su13042025**

Arogundade, O. R., & Palla, K. (2023). Virtualization revolution: Transforming cloud computing with scalability and agility.10.17148/IARJSET.2023.106104

Lal, R., Chhabra, A., Singla, S., & Sharma, D. (2024, April). Blockchain Technology: Revolutionizing Trust, Transparency, and Transaction Efficiency. In *2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS)* (Vol. 1, pp. 1-5). IEEE.10.1109/ICKECS61492.2024.10616743

Lang, V. (2021). Digitalization and digital transformation. In *Digital Fluency: Understanding the Basics of Artificial Intelligence, Blockchain Technology, Quantum Computing, and Their*

Applications for Digital Transformation (pp. 1-50). Berkeley, CA: Apress. https://doi.org/10.1007/978-1-4842-6774-5_1

Alsharari, N. (2021). Integrating blockchain technology with internet of things to efficiency. *International Journal of Technology Innovation and Management (IJTIM)*, *1*(2), 01-13.https://doi.org/10.54489/ijtim.v1i2.25

Wang, Z., Li, M., Lu, J., & Cheng, X. (2022). Business Innovation based on artificial intelligence and Blockchain technology. *Information Processing & Management*, *59*(1), 102759.https://doi.org/10.1016/j.ipm.2021.102759

Mu, X., & Antwi-Afari, M. F. (2024). The applications of Internet of Things (IoT) in industrial management: a science mapping review. *International Journal of Production Research*, *62*(5), 1928-1952.https://doi.org/10.1080/00207543.2023.2290229

Khan, Y., Su'ud, M. B. M., Alam, M. M., Ahmad, S. F., Ahmad, A. Y. B., & Khan, N. (2022). Application of internet of things (IoT) in sustainable supply chain management. *Sustainability*, *15*(1), 694.**https://doi.org/10.3390/su15010694**

Laurențiu-George, D. I. N. U. (2022). Using Cryptocurrencies, a Management Strategy for the Future. *Internal Auditing & Risk Management*, *65*(1), 19-32.

How, M. L., & Cheah, S. M. (2024). Forging the future: Strategic approaches to quantum ai integration for industry transformation. *Ai*, *5*(1), 290-323.**https://doi.org/10.3390/ai5010015**

How, M. L., & Cheah, S. M. (2023). Business Renaissance: Opportunities and challenges at the dawn of the Quantum Computing Era. *Businesses*, *3*(4), 585-605.**https://doi.org/10.3390/businesses3040036**

Ingale, A. (2024). IT Governance: Navigating the Digital Transformation Era. *IT governance, digital transformation, large-scale data analytics, cloud computing*. https://dx.doi.org/10.2139/ssrn.502836 1

Alyami, H., Nadeem, M., Alosaimi, W., Alharbi, A., Kumar, R., Gupta, B. K., ... & Khan, R. A. (2022). Analyzing the data of software security life-span: quantum computing era. *Intelligent Automation & Soft Computing*, *31*(2), 707-716.

Krings, W. (2025). Empowering Innovation: A Novel Startup in Educational Advisory. *Journal of Next-Generation Research 5.0*.

Block, S. (2023). Digital Transformation & Agile Prioritization. In *Large-Scale Agile Frameworks: Agile Frameworks, Agile Infrastructure and Pragmatic Solutions for Digital Transformation* (pp. 9-45). Berlin, Heidelberg: Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-67782-7_2

Kamariotou, M., & Kitsios, F. (2022). Hackathons for driving service innovation strategies: The evolution of a digital platform-based ecosystem. *Journal of Open Innovation: Technology, Market, and Complexity*, *8*(3), 111.https://doi.org/10.3390/joitmc8030111

Kuchuk, H., & Malokhvii, E. (2024). Integration of IoT with cloud, fog, and edge computing: a review. *Advanced Information Systems*, *8*(2), 65-78. http://ais.khpi.edu.ua/article/view/305471

How Do Agile Development Practices and Innovation-Driven Initiatives Like Hackathons Support Digital Transformation in Organizations? Discuss the Role of Cloud and Fog Computing in Facilitating This Change.

Agile development practices and innovation-driven initiatives such as hackathons have become crucial components in fostering **digital transformation** within organizations. As businesses strive to become more **digital-first**, they must adapt quickly to changing market demands, customer expectations, and technological advancements. Traditional methods of software development and innovation are often too slow or inflexible to keep pace with the rapid pace of change. Agile methodologies and innovation-driven events like hackathons enable organizations to be more adaptive, iterative, and collaborative. Moreover, the **cloud** and **fog computing** play vital roles in supporting this transformation by providing the necessary infrastructure and capabilities to facilitate agile workflows, enhance collaboration, and ensure the scalability required for digital innovations.

Agile Development Practices and Digital Transformation

Agile development refers to an approach to software development that emphasizes flexibility, collaboration, and rapid iteration. Unlike traditional **Waterfall** development, which follows a linear sequence of stages, Agile development breaks projects into smaller, manageable chunks called **sprints**. These sprints allow for continuous improvement and quick adaptation to changing requirements. Agile methodologies encourage iterative progress, regular feedback from stakeholders, and the rapid delivery of functional software.

Agile practices are essential for **digital transformation** because they help organizations stay ahead of the curve in a fast-evolving digital landscape. Digital transformation involves adopting new technologies, improving processes, and rethinking customer engagement to deliver greater value. Agile allows organizations to break down large, complex digital transformation initiatives into smaller, more manageable parts, enabling faster execution and better alignment with customer needs.

Scrum and **Kanban** are two common Agile frameworks that provide organizations with structured methods for managing workflows, prioritizing tasks, and delivering products incrementally. These frameworks focus on transparency, accountability, and continuous feedback, which helps ensure that the software or digital solution being developed meets user needs and business goals.

For example, an e-commerce company adopting digital transformation might use Agile practices to incrementally develop and launch new features for its website or mobile app, ensuring that the end product is aligned with customer expectations and can be quickly adjusted based on user feedback. Agile development empowers teams to deliver results faster and with greater accuracy, providing businesses with a competitive advantage in the digital economy.

Hackathons: Driving Innovation and Collaboration

Hackathons have emerged as a powerful tool for driving **innovation** and encouraging collaboration within organizations. A hackathon is typically a short, intense event where developers, designers, and other stakeholders come together to work on a specific problem or create new ideas in a rapid and highly collaborative environment. Hackathons are designed to encourage creative thinking, break down silos, and foster cross-functional collaboration—critical elements in driving **digital transformation**.

The primary goal of a hackathon is to rapidly prototype solutions or develop innovative ideas that can later be refined and implemented into the organization's digital strategy. Hackathons also encourage participants to experiment with new technologies and develop **proofs of concept** for potential digital products, services, or improvements to existing systems. This innovation-driven environment accelerates the development of creative solutions and helps organizations stay at the forefront of technological advancements.

Hackathons are particularly beneficial in fostering a culture of **collaboration** and **continuous learning**, which are key to successful digital transformation. By involving employees from different departments (e.g., marketing, IT, operations), hackathons enable cross-pollination of ideas, leading to more comprehensive solutions that better address organizational needs.

One example of a company leveraging hackathons for digital transformation is **Facebook**. The company regularly holds internal hackathons where employees are given the freedom to work on innovative projects that could potentially become part of the core business offerings. These initiatives have resulted in the development of products like **Facebook Messenger** and **Timeline**, which have become central to the platform's user experience.

The Role of Cloud and Fog Computing in Facilitating Digital Transformation

The success of Agile development practices and hackathons relies heavily on the technological infrastructure that supports **collaboration**, **scalability**, and **efficiency**. This is where **cloud computing** and **fog computing** come into play, offering the necessary resources to facilitate agile workflows and innovation-driven initiatives.

Cloud computing refers to the delivery of IT services and resources—such as computing power, storage, and software—over the internet. Cloud-based infrastructure allows organizations to access scalable, on-demand resources without having to maintain physical data centers or invest in expensive hardware. This level of flexibility and scalability is essential for Agile development, which requires quick access to resources and the ability to scale up or down depending on the project's needs.

Agile teams can leverage cloud platforms such as **Amazon Web Services (AWS)**, **Microsoft Azure**, or **Google Cloud** to quickly spin up virtual environments for development, testing, and deployment. These cloud platforms provide a range of tools, including development frameworks, databases, machine learning models, and AI services, which can be easily integrated into Agile workflows. The cloud also supports the **DevOps** approach, where development and operations teams collaborate to automate the software delivery pipeline and ensure continuous integration and continuous deployment (CI/CD).

Fog computing, on the other hand, is a decentralized computing infrastructure that extends the capabilities of cloud computing to the edge of the network. Unlike cloud computing, which processes data in centralized data centers, fog computing processes data closer to the **source of the data** (e.g., IoT devices, sensors, or local networks). This is particularly important for real-time applications that require low latency and high performance.

Fog computing plays a crucial role in supporting digital transformation initiatives that involve **IoT** and other edge-based technologies. For instance, in a manufacturing environment, **IoT sensors** installed on production equipment generate data that needs to be processed quickly to prevent equipment failure or optimize operations. By processing data at the edge through fog computing, organizations can respond to these events in real-time without the delay of sending data to a centralized cloud server.

A key example of fog computing in action is **Cisco**, which integrates fog computing into its **IoT solutions** to improve industrial automation, smart cities, and healthcare systems. Cisco's fog computing infrastructure allows businesses to collect, analyze, and act on data in real-time, enabling faster decision-making and more efficient operations.

Conclusion

Agile development practices and innovation-driven initiatives like hackathons are fundamental to driving **digital transformation** within organizations. These practices enable businesses to quickly adapt to changes, foster collaboration, and deliver innovative solutions that align with evolving market demands. The **cloud** and **fog computing** further enhance these efforts by providing scalable, flexible, and efficient infrastructures that support the needs of Agile teams and innovation-driven projects.

Cloud computing allows businesses to access resources on demand, providing the infrastructure needed to support rapid development cycles, while fog computing extends the capabilities of cloud-based solutions by enabling real-time data processing at the edge. Together, these technologies create an environment where businesses can experiment, innovate, and rapidly implement solutions that drive digital transformation and competitive advantage in the marketplace.

As organizations continue to embrace Agile practices and hackathons, coupled with the power of cloud and fog computing, they will be better equipped to navigate the challenges of the digital era and maintain a competitive edge in an increasingly digital world.

How Do Agile Development Practices and Innovation-Driven Initiatives Like Hackathons Support Digital Transformation in Organizations? Discuss the Role of Cloud and Fog Computing in Facilitating This Change.

Agile development practices and innovation-driven initiatives such as hackathons have become crucial components in fostering **digital transformation** within organizations. As businesses strive to become more **digital-first**, they must adapt quickly to changing market demands, customer expectations, and technological advancements. Traditional methods of software development and innovation are often too slow or inflexible to keep pace with the rapid pace of change. Agile methodologies and innovation-driven events like hackathons enable organizations to be more adaptive, iterative, and collaborative. Moreover, the **cloud** and **fog computing** play vital roles in supporting this transformation by providing the necessary infrastructure and capabilities to facilitate agile workflows, enhance collaboration, and ensure the scalability required for digital innovations.

Agile Development Practices and Digital Transformation

Agile development refers to an approach to software development that emphasizes flexibility, collaboration, and rapid iteration. Unlike traditional **Waterfall** development, which follows a linear sequence of stages, Agile development breaks projects into smaller, manageable chunks called **sprints**. These sprints allow for continuous improvement and quick adaptation to changing requirements. Agile methodologies encourage iterative progress, regular feedback from stakeholders, and the rapid delivery of functional software.

Agile practices are essential for **digital transformation** because they help organizations stay ahead of the curve in a fast-evolving digital landscape. Digital transformation involves adopting new technologies, improving processes, and rethinking customer engagement to deliver greater value. Agile allows organizations to break down large, complex digital transformation initiatives into smaller, more manageable parts, enabling faster execution and better alignment with customer needs.

Scrum and **Kanban** are two common Agile frameworks that provide organizations with structured methods for managing workflows, prioritizing tasks, and delivering products incrementally. These frameworks focus on transparency, accountability, and continuous feedback, which helps ensure that the software or digital solution being developed meets user needs and business goals.

For example, an e-commerce company adopting digital transformation might use Agile practices to incrementally develop and launch new features for its website or mobile app, ensuring that the end product is aligned with customer expectations and can be quickly adjusted based on user feedback. Agile development empowers teams to deliver results faster and with greater accuracy, providing businesses with a competitive advantage in the digital economy.

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Hackathons have emerged as a powerful tool for driving **innovation** and encouraging collaboration within organizations. A hackathon is typically a short, intense event where developers, designers, and other stakeholders come together to work on a specific problem or create new ideas in a rapid and highly collaborative environment. Hackathons are designed to encourage creative thinking, break down silos, and foster cross-functional collaboration—critical elements in driving **digital transformation**.

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The Role of Cloud and Fog Computing in Facilitating Digital Transformation

The success of Agile development practices and hackathons relies heavily on the technological infrastructure that supports **collaboration**, **scalability**, and **efficiency**. This is where **cloud computing** and **fog computing** come into play, offering the necessary resources to facilitate agile workflows and innovation-driven initiatives.

Cloud computing refers to the delivery of IT services and resources—such as computing power, storage, and software—over the internet. Cloud-based infrastructure allows organizations to access scalable, on-demand resources without having to maintain physical data centers or invest in expensive hardware. This level of flexibility and scalability is essential for Agile development, which requires quick access to resources and the ability to scale up or down depending on the project's needs.

Agile teams can leverage cloud platforms such as **Amazon Web Services (AWS)**, **Microsoft Azure**, or **Google Cloud** to quickly spin up virtual environments for development, testing, and deployment. These cloud platforms provide a range of tools, including development frameworks, databases, machine learning models, and AI services, which can be easily integrated into Agile workflows. The cloud also supports the **DevOps** approach, where development and operations teams collaborate to automate the software delivery pipeline and ensure continuous integration and continuous deployment (CI/CD).

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